COURSE SYLLABUS

Data Analytics

24 modules | 96 sessions | live classes, recordings, assignments and placement support

01. Orientation and the analyst role (Week 1)

- What data analysts actually do day to day
- Tools we will use: Excel, SQL, Python, Power BI
- Analytics vs data science vs engineering
- Setting up your learning environment

02. Spreadsheet foundations (Week 1)

- Formulas, references and named ranges
- Sorting, filtering and conditional formatting
- Pivot tables and pivot charts
- Data cleaning in spreadsheets

03. Business metrics and KPIs (Week 2)

- Revenue, margin, CAC, LTV and churn
- Choosing metrics that match a business goal
- Defining a metric precisely
- Building a KPI sheet

04. Data types and data quality (Week 2)

- Categorical, numeric, temporal and text data
- Duplicates, outliers and missing values
- Validation rules and source checks
- Documenting a dataset

05. SQL foundations (Week 3)

- Tables, schemas and data types
- SELECT, WHERE, ORDER BY and LIMIT
- Aggregations and GROUP BY
- CASE statements and derived columns

06. SQL joins and subqueries (Week 4)

- Inner, left, right and full joins
- Self joins and relationship patterns
- Subqueries and CTEs
- Window functions for ranking and running totals

07. SQL for analysis (Week 4)

- Cohort and retention queries
- Funnel analysis in SQL
- Time series aggregation
- Query performance and indexes

08. Python for analysts (Week 5)

- Python basics in a notebook
- Lists, dicts and loops for data work
- Functions and reusable analysis steps
- Debugging and reading errors

09. Pandas essentials (Week 5)

- Series and DataFrames
- Selecting, filtering and sorting

- Groupby, pivot and merge
- Cleaning real messy data

10. Exploratory data analysis (Week 6)

- Summary statistics and distributions
- Missing value strategies
- Correlation and relationships
- Asking the next question

11. Visualisation principles (Week 6)

- Chart selection and misleading charts
- Colour, labels and annotation
- Matplotlib and Seaborn basics
- Designing a one page narrative

12. Statistics for decisions (Week 7)

- Mean, median, variance and percentiles
- Sampling and confidence intervals
- Hypothesis testing and p values
- A/B test design and reading results

13. Excel to dashboard workflow (Week 7)

- Preparing a clean model sheet
- Power Query style transformations
- Chart design in Excel
- Saving and sharing a report pack

14. Power BI foundations (Week 8)

- Importing data and the data model
- Relationships and star schema
- DAX basics: measures and calculated columns
- Building report pages

15. Power BI advanced (Week 8)

- Time intelligence measures
- Filters, slicers and drill through
- Row level security
- Publishing to the Power BI service

16. Tableau essentials (Week 9)

- Connecting data and building views
- Dimensions, measures and aggregations
- Calculated fields and parameters
- Dashboards and actions

17. Dashboard storytelling (Week 9)

- Defining the audience and the decision
- Layout hierarchy and visual weight
- Annotations that explain the why
- Review checklist before publishing

18. Data cleaning at scale (Week 10)

- Deduplication strategies
- Standardising names, dates and categories
- Joining sources with different keys
- Automating a repeatable cleaning pipeline

19. Advanced Excel analytics (Week 10)

- Lookup patterns with XLOOKUP and INDEX MATCH
- What if analysis and goal seek
- Pivot based mini dashboards
- Macros and VBA introduction

20. Forecasting and trends (Week 11)

- Moving averages and trend lines
- Seasonality and decomposition
- Basic regression for forecasting
- Communicating uncertainty

21. AI assisted analysis (Week 11)

- Using AI to draft queries and explanations
- Validating AI output against the data
- Automating summaries and insights
- Privacy and data handling rules

22. Portfolio project: sales analytics (Week 12)

- Business brief and questions
- SQL extracts and cleaning
- Dashboard build and storytelling
- Findings deck and recommendations

23. Portfolio project: marketing funnel (Week 12)

- Channel level data model
- Funnel and cohort analysis
- Cost and ROI reporting
- Stakeholder presentation

24. Interview and case preparation (Week 13)

- SQL interview question bank
- Case study frameworks
- Explaining dashboard decisions
- Mock interview and feedback

Fees, batch dates and mode (online / classroom): confirm the current numbers on the website or with the admissions team before enrolling. Yantrix Technology, Pune.